Global Perception Network for Salient Object Detection in Remote Sensing Images

Yu Liu, Shanwen Zhang, Zhen Wang, *Student Member, IEEE*, Baoping Zhao, and Lincheng Zou

***Abstract*— Despite recent works that have achieved remarkable progress on salient object detection for natural scene images, to detect various types and scales of objects, complex backgrounds in remote sensing images are still challenging. In this study, a novel global perception network (GPNet) is constructed for the salient object detection of remote sensing images. The proposed GPNet includes a global perception module (GPM), an axial attention block (AAB), and a feature distillation structure (FDS). The GPM is used to preserve the relationships of the entire dataset, the AAB is designed to capture the dependencies between the space and channel, the FDS is introduced to enable the helpful multilevel information flow into deep layers to enhance feature generation, and the global and the local attention information are mutually fused to enhance the network mode. Extensive experiments on three public datasets demonstrate that the proposed method outperforms other compared state-of-the-art methods both qualitatively and quantitatively (https://github.com/liuyu1002/GPnet).**

***Index Terms*— Feature distillation structure (FDS), global perception module (GPM), remote sensing images, salient object detection.**

I. Introduction

SALIENT object detection aims to extract the most conspicuous and eye-catching objects in an image. Salient object detection can be viewed as a preprocessing process in many computer vision tasks, including visual tracking [1], image retrieval [2], quality assessment [3], and image segmentation [4]. Salient object detection has highly practical value in remote sensing images and can be applied in many visual and military applications, such as object extraction [5], image fusion [6], and oil tank detection [7]. Many salient object detection methods have been presented based on various low-level handcrafted features [8]. Its performance mainly relies on the extracted handcrafted features that are subjective and have poor generalization. However, not all the extracted features contribute to the saliency detection in an image, and there are some distractions from complex backgrounds.

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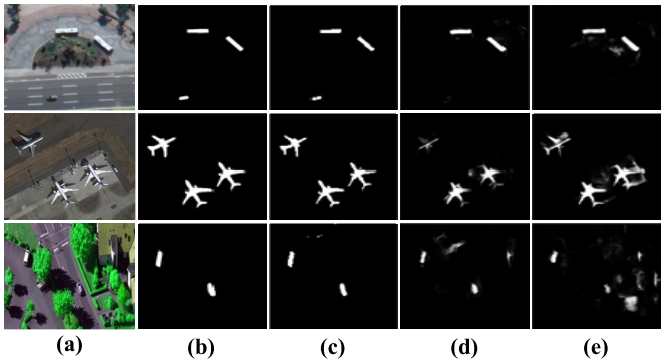


Fig. 1. Visualization of the saliency maps of different methods. (a) Remote sensing images. (b) Ground truth. (c) Our method. (d) PAGR [13]. (e) ASNet [28].

Along with the increase in deep learning methods, convolutional neural networks (CNNs) and their modified methods have achieved remarkable performance in salient object detection [9]. However, remote sensing images are irregularly collected from a high angle by aerial sensors and satellites. It is difficult to directly extend the success of salient object detection for natural scene images to optical remote sensing images because the actual shapes and scales of objects in remote sensing images usually vary.

Conventional layer-to-layer models cannot effectively detect salient objects due to the local receptive field in convolution operations, and not all channels focus on salient objects [10], [11]. Thus, various attention mechanisms have been proposed to capture spatial and channel relationships [12], [13]. The pyramid attention mechanism has been widely used to enhance the representation ability of a model and emphasizes the importance of salient edge information [14]. However, these methods concentrate on the spatial or channel relationships in a single sample and neglect the relationships in the overall sample (see the fourth column in Fig. 1). Global relationships are vital for the saliency detection of remote sensing images because the salient objects in remote sensing images usually vary.

To alleviate the interference of scale variability, the existing state-of-the-art methods based on deep CNNs aim to integrate multilevel features. Hou *et al.* [15] introduced short connections within the holistically nested edge detector (HED) to integrate different level features. Zhang *et al.* [12] proposed a dense connection structure to integrate low- and

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high-level attention features. Zhang *et al.* [16] designed a bidirectional structure to pass information between multilevel features and used a gate function to control message passing. Pang *et al.* [17] proposed a feature aggregation structure to integrate features on adjacent levels.

The above methods simply extract the features from the convolutional layer, which may lead to distraction due to the limitation of the convolution operation (the fifth column in Fig. 1). In this study, the global perception network (GPNet) is proposed for the salient object detection of remote sensing images. In GPNet, the global attention mechanism is used to capture the relationships in the overall dataset, and the axial attention block (AAB) is designed to obtain both channel and spatial dependencies. Then, the global and local attention information is mutually fused to better alleviate the interference of object variation and complex backgrounds. Moreover, the feature distillation structure (FDS), which can filter the useless multilevel information and allow the helpful multilevel information to flow into deeper layers to guide the generation of features, is constructed.

The main contributions of this study are summarized as follows.

1) A novel salient object detection model called GPNet is constructed. Different from previous works, global attention information is used to gradually enhance the feature representation.
2) The global perception module (GPM) and AAB are designed to obtain global and local attention relationships, and the relationships can be fused to suppress the interference of object variation and complex backgrounds.
3) The FDS is constructed to extract information from multilevel features efficiently, making the saliency maps more discriminative.

The remainder of this article is arranged as follows. Section II reviews the related works. Section III introduces the details of the proposed GPNet. Section IV conducts extensive experiments to demonstrate the effectiveness of the proposed methods, and finally, Section V concludes this article.

II. Related Work

A. Salient Object Detection

Traditional salient object detection methods are based on handcrafted features, such as contrast [18]–[20], boundary background [21], [22], color space [23], center prior [24], and visual attention [25], [26]. Borji *et al.* [27] extensively compared 41 state-of-the-art models qualitatively and quantitatively on seven challenging datasets.

Unlike handcrafted methods, CNNs can automatically extract high-level semantic features for specific vision tasks. CNN-based methods have remarkably improved salient object detection tasks [14]–[17], [28]–[37]. Deng *et al.* [29] proposed an attentive saliency network to learn salient objects from fixation maps, which captures a high-level understanding of the scene. Liu *et al.* [30] introduced a feature pyramid network with two novel pooling-based modules for real-time saliency detection. Su *et al.* [32] used boundary localization and integrated successive dilation module streams to enhance feature selectivity and invariance.

B. Salient Objection Detection for Optical Remote Sensing Images

Salient object detection for remote sensing images is an important task that can be widely applied in many related remote sensing image processing tasks, such as building extraction [38], harbor detection [39], ship detection [40], and airport detection [41]. Liu *et al.* [26] introduced a self-adaptive multiple feature fusion method and adopted integrating color, intensity, texture, and global contrast features to produce feature maps. Li *et al.* [42] proposed the first deep learning-based approaches, consisting of a two-stream pyramid module and encoder and decoder modules. This work also provided a publicly available dataset for remote sensing saliency detection. Li *et al.* [37] combined different levels and resolution features to locate salient objects. Zhang *et al.* [12] designed a dense attention fluid network consisting of a global context-aware attention module for capturing context relationships and constructed a dense attention fluid structure for better information transmission. Tu *et al.* [43] proposed a multiscale joint region and boundary model to capture the multiscale region feature of salient objects. Zhou *et al.* [44] used the edge information to locate salient objects.

C. Attention Mechanisms

Recently, attention mechanisms have shown efficiency in various vision tasks, such as object detection [12], [33], [35], image classification [45], and image segmentation [46]. Hu *et al.* [45] adopted a pooling operation for each feature map and used a fully connected layer to capture the channel relationships. Hou *et al.* [47] proposed a coordinate attention mechanism for mobile networks by embedding spatial information into channel information. Woo *et al.* [48] proposed a convolutional block attention module, which sequentially captures attention maps along the spatial and channel dimensions. Misra *et al.* [49] proposed a triplet attention mechanism that captures attention weights using a three-branch structure. However, these methods have difficulty in capturing rich context or channel information due to the pooling operation.

Nonlocal attention mechanisms can capture the long-range dependencies by encoding all the interpixel relationships and have been widely applied to many computer vision tasks [50]. Fu *et al.* [46] proposed a dual attention network to integrate local features with global dependencies. Zhang *et al.* [12] proposed a global feature aggregation mechanism that captures spatial information. Zhang *et al.* [13] further introduced both spatial and channel attention mechanisms for saliency detection. Feng *et al.* [35] proposed an attentive feedback module to extract the information of objects better. However, these methods only capture the dependencies in one sample while overlooking the relationships in the entire dataset.

D. Multilevel Feature

In recent years, several successful applications of feature aggregation from multiple layers have proven

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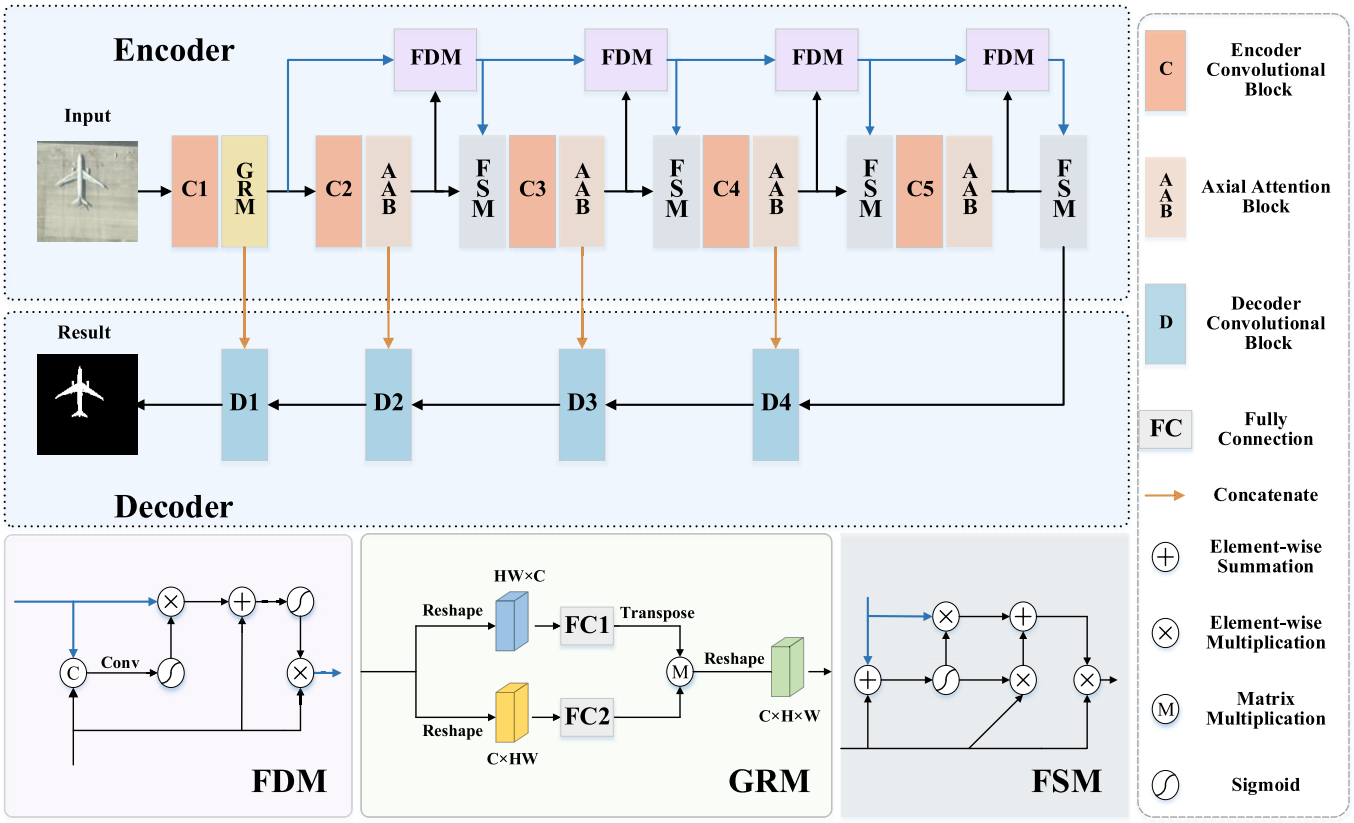


Fig. 2. Overall architecture of our proposed GPNet, which is an encoder–decoder style. The encoder consists of: 1) the backbone encoder convolutional blocks to extract multilevel features; 2) a GPM to capture global relationships in the entire dataset; 3) the AABs to obtain spatial and channel dependencies; 4) the FDM to aggregate the previous useful information; and 5) the FSM to filter the redundant information in backbone convolutional blocks. The decoder consists of backbone decoder convolutional blocks. The concatenate operation is used to fuse features on the same level between the encoder and decoder convolutional blocks to enhance the feature representation. The conv in the FDM represents the convolutional operation.

their effectiveness. Zhang *et al*. [51] introduced an aggregating multiple features framework, which uses the combination of all levels of features to predict saliency maps. Zhang *et al.* [16] designed a bidirectional structure that uses a gate function to control the information passing between different levels. Pang *et al.* [17] proposed an aggregate interaction module to integrate adjacent level features. These methods may cause information redundancy due to coarse fusion or direct aggregation from convolutional feature maps.

III. Proposed Method

A. Overview

Inspired by the above methods, a novel attention mechanism called the GPM is proposed to preserve the global information in the entire dataset. The AAB is proposed to capture spatial and channel relationships. Unlike the above nonlocal-based approaches, the AAB is more efficient at capturing spatial information and channel relationships by computing the input features along the horizontal and vertical directions. Moreover, the FDS is constructed to filter the redundant multilevel information to enhance information propagation. Based on the GPM, AAB, FDS, and UNet [11], the GPNet is constructed. It is an encoder–decoder structure, and its overall architecture is shown in Fig. 2. Considering the roughness of directly using features in sequence and the lack of interaction relationships between the pixel-wise features in UNet, the GPM is added after the first convolutional block to capture the relationships in a dataset, and each convolutional block other than the first block in the encoder path is equipped with an AAB to capture both spatial and channel dependencies. The FDS consists of a feature distillation module (FDM) and a feature selection module (FSM). Each FDM receives features generated from the corresponding AAB and previous FDM (GPM if none) to produce multilevel features, which are further added to the input features through a residual connection to formulate the enhanced features. Each FSM receives the enhanced features generated from the corresponding FDM and attentive features from the previous AAB to guide feature propagation. During the decoder network, the homologous levels of features are concatenated to generate saliency maps.

B. Global Perception Module

The diversity of the salient objects and the complexity of the backgrounds are the two significant challenges for the salient object detection of remote sensing images. Thus, the capability to distinguish features is an essential component in detection. Recently, the attention mechanism has been adopted in salient object detection to capture long-range dependencies [4], [14], [30]. However, these attention mechanisms concentrate on the relationships between different locations in

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a single input, which may lead to incomplete saliency detection for remote sensing images. To this end, the GPM is proposed to capture the affinities across the entire dataset, which is beneficial to generating discriminative features for saliency prediction. The GAM consists of two external memory units. Specifically, the two external memory units are implemented as linear layers, which can be optimized during the training stage.

As shown in Fig. 2, an input $f^1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ is reshaped to $\mathbb{R}^{C_1 \times N_1}$, where $N_1 = H_1 \times W_1$. Then, the relationship maps $A_s \in \mathbb{R}^{C_1 \times k}$ and $A_c \in \mathbb{R}^{N_1 \times k}$ between the feature and two external memory units $M_h \in \mathbb{R}^{N_1 \times k}$ and $M_c \in \mathbb{R}^{C_1 \times k}$, where k is a hyperparameter, are, respectively, computed. The relationship maps are calculated as follows:

$$A_s = f^1 M_h \tag{1}$$

$$A_c = \left(f^1\right)^T M_c \tag{2}$$

where $(f^1)^T$ is the transpose of f^1.

Then, the global attention map $G \in \mathbb{R}^{C_1 \times N_1}$ is calculated as follows:

$$G = A_s A_c^T \tag{3}$$

where A_c^T is the transpose of A_c. Then, we reshape G to $\mathbb{R}^{C_1 \times H_1 \times W_1}$.

Finally, the final output feature $R \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ is realized by a residual connection

$$R = f^1 + \delta \cdot G \tag{4}$$

where δ denotes a learnable parameter to control the contribution of the global attention influences.

In this way, the external linear layers can store the interpixel information of the input features and adaptively consider the entire dataset. Thus, the different sample information interacts to improve saliency detection.

C. Axial Attention Block

Spatial information is critical for a model to learn discriminant feature representations, and explicit channel relationships can increase the model's sensitivity for the class-specific response. However, due to the local receptive field of conventional convolution operations, the learned feature belonging to the same object may lead to inconsistency. To this end, the self-attention mechanism is proposed to capture the long-range dependencies in vision tasks. However, previous works [12], [46], [52] often used the position or channel self-attention mechanism separately, thus neglecting the interleaving information of positions and channels. To this end, an efficient AAB inspired by the coordinate attention mechanism is proposed [47]. The AAB can integrate both spatial dependencies and channel relationships, which are beneficial to locating salient objects and obtaining more explicit boundaries simultaneously. In AAB, the position and channel information can be encoded in three ways: horizontal attention embedding, vertical attention embedding, and attention aggregation.

1) Horizontal Attention Module: A horizontal attention module is designed to embed the spatial dependencies into channel information. Specifically, the input $f^s \in \mathbb{R}^{C_s \times H_s \times W_s}$ is reshaped to $\mathbb{R}^{P_s \times N_s}$, where $s \in \{2, 3, 4, 5\}$ and $N_s = H_s \times W_s$. Then, matrix multiplication between f^s and the transpose of f^s is performed to obtain the horizontal correlation maps $H^s \in \mathbb{R}^{P_s \times P_s}$, and the horizontal relationship maps $K^s \in \mathbb{R}^{P_s \times P_s}$ are calculated as follows:

$$k_{ij} = \frac{\exp\left(h_{ij}^s\right)}{\sum_{i=1}^{P_s} \exp\left(h_{ij}^s\right)} \tag{5}$$

where h_{ij}^s represents the entry at matrix H^s at location (i, j) and h_{ij}^s measures the impact of k_i on k_j.

In this way, the horizontal interpixel information is encoded in K^s. Then, matrix multiplication between K^s and f^s is conducted to generate horizontal attention feature D^s and reshape it to $\mathbb{R}^{C_s \times H_s \times W_s}$. Finally, the final output $F^s \in \mathbb{R}^{C_s \times H_s \times W_s}$ is obtained using a residual connection as follows:

$$F^s = \delta \cdot D^s + f^s \tag{6}$$

where δ denotes a learnable parameter. In this way, each entry of output F^s encodes the horizontal dependencies.

2) Vertical Attention Module: In order to capture vertical dependencies, a vertical attention module is built. It is worth noting that our vertical attention module and horizontal attention module are complementary. Different from the horizontal attention module, $f^s \in \mathbb{R}^{C_s \times H_s \times W_s}$ is reshaped to $\mathbb{R}^{L_s \times W_s}$, where $s \in \{2, 3, 4, 5\}$ and $L^s = C_s \times H_s$. Then, matrix multiplication between f^s and the transpose of f^s is performed to generate a vertical correlation map $M^s \in \mathbb{R}^{L_s \times L_s}$. Then, the vertical relationship maps $V^s \in \mathbb{R}^{L_s \times L_s}$ are calculated as follows:

$$v_{ij} = \frac{\exp\left(m_{ij}^s\right)}{\sum_{i=1}^{L_s} \exp\left(m_{ij}^s\right)} \tag{7}$$

where m_{ij}^s denotes the entry at location (i, j) of matrix M^s and v_{ij} measures the impact of position m_i on position m_j.

Then, matrix multiplication between V^s and f^s is performed to obtain $O^s \in \mathbb{R}^{L_s \times W_s}$ and reshape it to $\mathbb{R}^{C_s \times H_s \times W_s}$, and the residual connection is used to obtain the final output $E^s \in \mathbb{R}^{C_s \times H_s \times W_s}$

$$E^s = \delta \cdot O^s + f^s \tag{8}$$

where δ denotes a learnable parameter, and each entry of output E^s is weighted by all vertical positions to suppress the distraction of backgrounds.

D. Feature Distillation Structure

The features of different layers in the CNN correspond to the different levels of semantic information. The features in the shallower layers capture spatial textures, while the deep features encode the high-level abstract semantics. The features of each layer contain essential information for salient object detection. Thus, the FDS is proposed to take full advantage of the features of different levels and filter redundant information. In this structure, the FDM is built to aggregate previous helpful information. The FSM is introduced to select the features that

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are saved and the features that are abandoned. In this way, the valuable cues of all the previous layers could propagate to the next layer and hence enhance the feature representation.

1) Feature Distillation Module: The details of the FDM are shown in Fig. 2. Using an AAB to produce an attention map V^s, where V^1 represents the feature map generated by the GPM, the process of generating distilled feature D^s is denoted as follows:

$$D^s = V^s \odot \sigma\left(V^s + V^{s-1} \odot \sigma\left(\text{conv}\left(\text{cat}\left(V^s, V^{s-1}\right)\right)\right)\right) \quad (9)$$

where $s \in \{1, 2, 3, 4, 5\}$, cat() denotes the concatenation operation along the channel dimension, conv() is the convolution operation, and σ and $\odot$ represent the sigmoid function and element-wise multiplication, respectively.

2) Feature Selection Module: The structure of the proposed FSM is shown in Fig. 2. Given the distilled feature D^s and feature V^s generated by the axial attention module (AAM), the selected feature E^s is formulated as follows:

$$E^s = \left(D^s + V^s\right) \odot \sigma\left(D^s + V^s\right) \quad (10)$$

where $s \in \{2, 3, 4, 5\}$, σ represents the sigmoid function, and $\odot$ denotes element-wise multiplication.

Then, the feature E^s is integrated into V^s with a residual connection to update V^s.

E. Loss Function

Unlike some recent saliency models trained with additional supervision constraints (i.e., edges [12], [14], [34], [36] and boundaries [35]), which need additional processes, we simply use the balanced binary cross-entropy loss function. Given a predicted saliency result S_m and the corresponding ground truth G_m, the loss function can be formulated as follows:

$$\varsigma_{\text{bce}} = -\left[\beta G_m \log(S_m) + (1-\beta)(1 - G_m)\log(1 - S_m)\right] \quad (11)$$

where $\beta = y_+/y_-$ is the loss weight. Here, y_+ and y_- are the foreground and background label sets, respectively.

IV. Experiments

A. Datasets

To evaluate the performance of the proposed method, we conduct extensive experiments on three public datasets, including the ORSSD [42], EORSSD [12], and ORSI-4199 dataset [43].

1) ORSSD Dataset: ORSSD is a publicly available dataset with pixel-wise ground truths for salient object detection in remote sensing images. It contains 800 images captured from Google Earth and existing remote sensing image datasets and includes 600 images in the training set and 200 images in the test set.

2) EORSSD Dataset: EORSSD is an extended dataset of ORSSD containing 2000 images and the corresponding high-quality pixel-level annotations. It has 1400 training images and 600 test images.

3) ORSI-4199 Dataset: ORSI-4199 is a challenging dataset with various sizes and scales of objects and background interference. It has 4199 images and corresponding pixel-level annotations.

B. Evaluation Metrics

To evaluate the performance of the different methods, precision–recall (PR) curves, mean absolute error (MAE), F-measure, and S-measure [53] are adopted.

1) PR Curves: A binary mask is obtained by binarizing the saliency map with a threshold sliding from 0 to 255. The precision and recall scores are further calculated by comparing the binary mask with the ground truth

$$\text{Precision} = \frac{|\text{SF} \cap \text{GF}|}{|\text{SF}|} \quad (12)$$

$$\text{Recall} = \frac{|\text{SF} \cap \text{GF}|}{|\text{GF}|} \quad (13)$$

where SF represents the pixels classified as the foreground and GF represents the pixels that belong to the foreground in the ground truth.

The PR curve can be drawn with the vertical axis denoting the precision score and the horizontal axis denoting the recall score.

2) Mean Absolute Error: The MAE score is used to measure the average difference between saliency map S and ground truth G at the pixel level, and it is calculated as follows:

$$\text{MAE} = \frac{1}{w \times h} \sum_{i=1}^{w} \sum_{j=1}^{h} |\, S(i, j) - G(i, j)| \quad (14)$$

where w and h represent the width and height of the output image, respectively.

3) F-Measure: Since high precision and recall are required in many tasks, the F-measure is defined as a harmonic mean of precision and recall and used to measure the overall performance

$$F_\beta = \frac{\left(1 + \beta^2\right) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} \times \text{Recall}} \quad (15)$$

where β^2 is set to 0.3 to highlight the precision.

4) S-Measure: The S-measure is used to evaluate the structural similarity between the ground truth and the saliency map

$$S_m = \alpha \times S_0 + (1 - \alpha) \times S_r \quad (16)$$

where α is set to 0.5 to denote equal contributions of both the region and object, as suggested in [53].

C. Implementations

The experiments are conducted on a workstation equipped with an Intel Xeon Silver 4210 CPU @ 2.20 GHz, 64-GB RAM, and an NVIDIA TITAN RTX 24-GB GPU; the Ubuntu 20.04 operating system; and the deep learning open-source library of PyTorch as the backend deep learning engine [54]. During the training process, the training set is divided into multiple minibatches, where the size of each minibatch is set to 8. The "Kaiming" method is used to initialize the filter weights of each layer [55]. All experiments are conducted using the ADAM optimizer with a base learning rate of 0.0001, a weight decay of 0.0005, and a momentum of 0.9 [56], and the "cosine annealing" strategy with $T_{\max} = 2000$ and $\text{eta}_{\min} = 0.00001$ is used. The number of iterations is set to 2000. In the GPM

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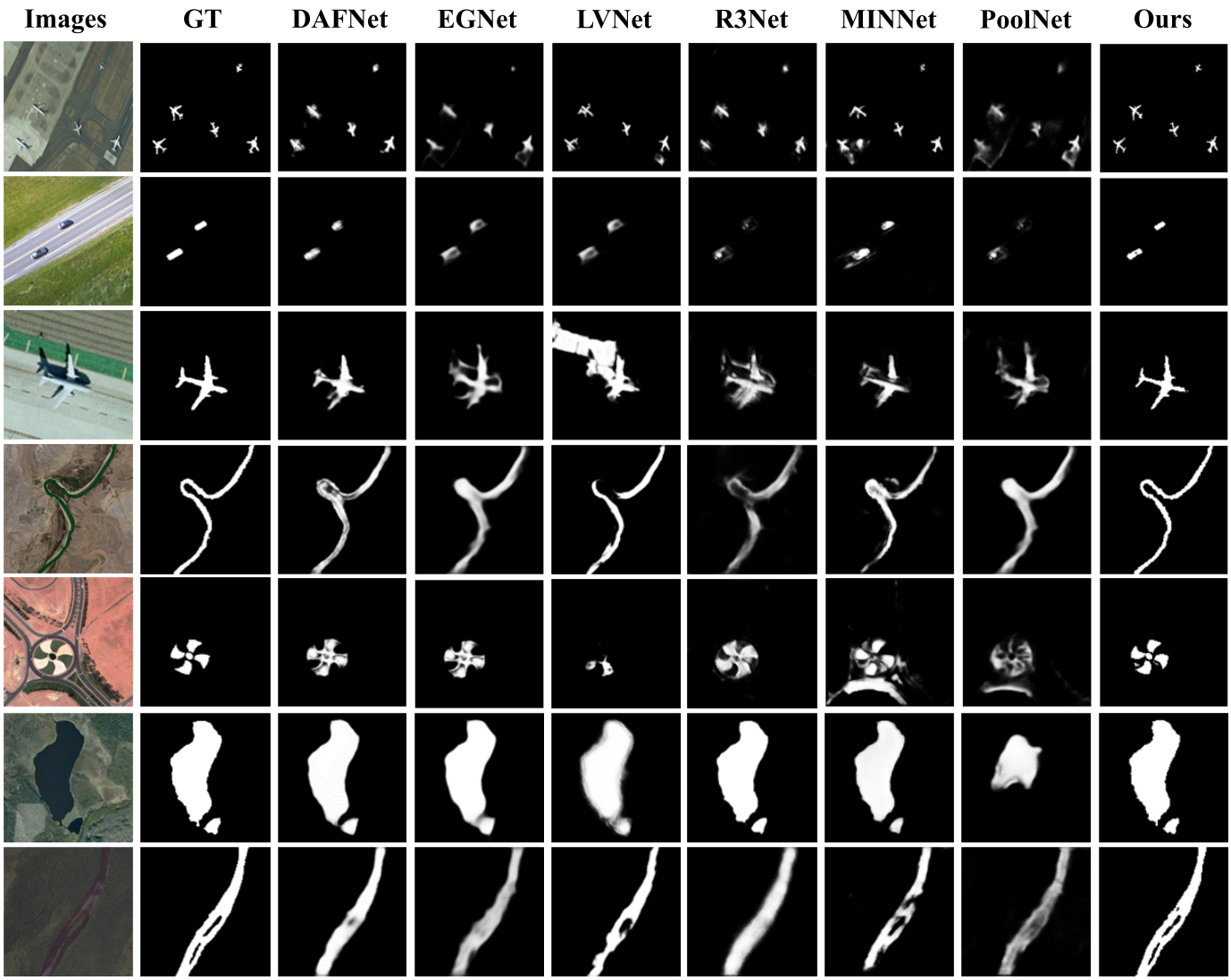


Fig. 3. Visualization of the saliency maps obtained by our method, DAFNet [12], EGNet [34], LVNet [42], R3Net [29], MINNet [17], and PoolNet [30]. It is obvious that our method achieves the best results.

and AABs, the learnable parameter is initialized as 0.1. Data augmentation techniques involving rotation and flipping are utilized to augment the training set, and then, each image is resized to 128×128 due to the limitation of the GPU memory.

D. Comparison With State-of-the-Art Methods

To evaluate the effectiveness of the proposed GPNet, we compare it with 12 state-of-the-art methods (i.e., ASNet [28], R3Net [29], BMPS [16], MINNet [17], POOLNet [30], CASNet [31], BANet [32], R2Net [33], EGNet [34], PAGR [13], LVNet [3], and DAFNet [4]). All the saliency maps of the other compared methods are generated by their source code.

1) Visual Comparisons: The visual comparisons of the proposed method with other compared methods are shown in Fig. 3. The images, including the aircraft, river, cars, pond, and buildings, are selected from the EORSSD dataset. Fig. 3 shows that the proposed methods achieve better background suppression and content integrity.

Specifically, in the first image, there are multiple aircraft, and the aircraft are relatively small. Thus, other compared methods cannot completely detect them. In the fifth image, two cars should be detected. However, the other compared methods fail to accurately and distinctly detect both cars. In contrast, our method can completely detect small objects. Moreover, our proposed method exhibits superiority in alleviating the interference of complicated backgrounds. In the sixth image, all the compared methods generate blurred responses with shallow backgrounds for the image. Our proposed method can effectively suppress the interferences of the shadow area. For the challenging scenarios in Fig. 3 (Rows 4–7), our method can still highlight the entire salient object with better structural integrity.

Due to our elaborately designed method, the salient maps generated by our method have more high-quality semantic information. Compared with other state-of-the-art methods, the benefits of our global perception structure and attention mechanisms are critical for detecting salient objects accurately.

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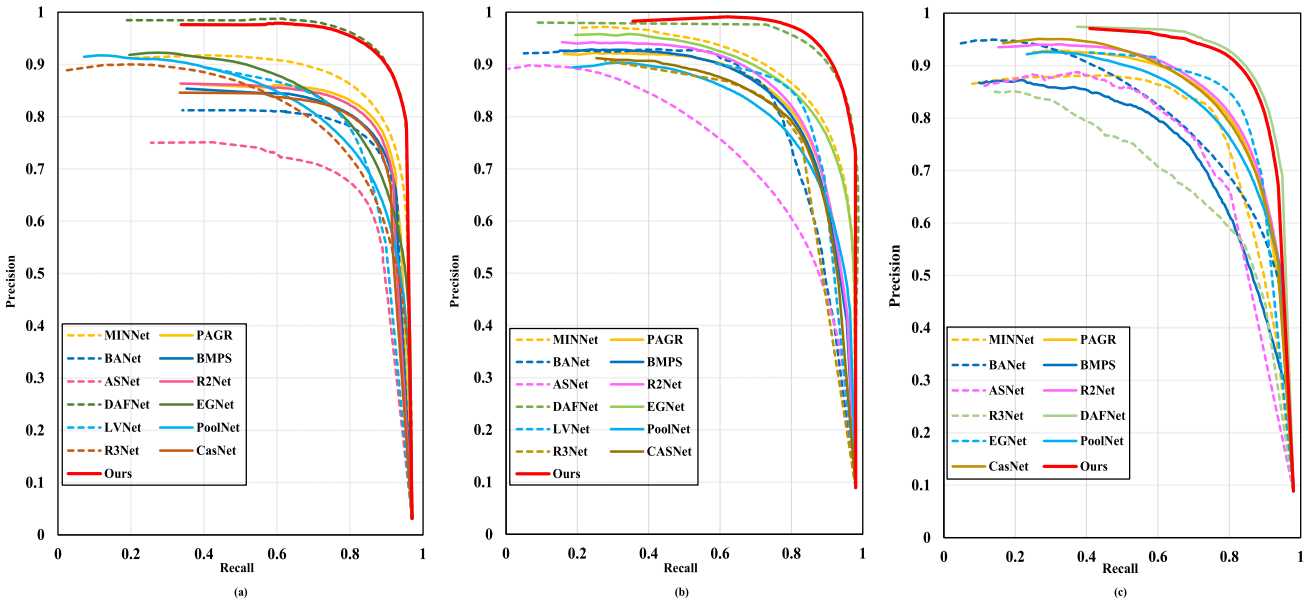


Fig. 4. Comparison of the different methods on three datasets by using PR curves. (a) Evaluation on the EORSSD dataset. (b) Evaluation on the ORSSD dataset. (c) Evaluation on the ORSI-4199 dataset.

TABLE I

QUANTITATIVE COMPARISONS IN TERMS OF THE MAE (SMALLER IS BETTER), F-MEASURE (LARGER IS BETTER), AND S-MEASURE (LARGER IS BETTER). THE TOP THREE RESULTS ARE MARKED IN RED , BLUE , AND GREEN , RESPECTIVELY

Method	ORSSD			EORSSD			ORSI-4199		
	MAE	F_b -	S_m -	MAE	F_b -	S_m -	MAE	F_b -	S_m -
ASNet	0.0582	0.7215	0.7952	0.0475	0.7615	0.7915	0.0432	0.8138	0.8306
PoolNet	0.0358	0.7911	0.8721	0.0209	0.7812	0.8218	0.0378	0.8401	0.8628
LVNet	0.0207	0.8414	0.8815	0.0145	0.8051	0.8645	-	-	-
MINNet	0.0145	0.8515	0.8982	0.0095	0.8625	0.8742	0.0405	0.8315	0.8569
PAGR	0.0301	0.8120	0.8315	0.0101	0.8354	0.8651	0.0383	0.8547	0.8672
BANet	0.1123	0.7505	0.7834	0.0124	0.7982	0.8615	0.0387	0.8209	0.8369
BMPS	0.0242	0.8189	0.8511	0.0214	0.7968	0.8343	0.0423	0.8026	0.8153
R2Net	0.0305	0.8124	0.8465	0.0151	0.8305	0.8612	0.0384	0.8521	0.8594
R3Net	0.0399	0.7998	0.8141	0.0171	0.7709	0.8193	0.0491	0.7876	0.8052
CASNet	0.0345	0.8109	0.8341	0.0212	0.7920	0.8304	0.0372	0.8469	0.8705
EGNet	0.0216	0.8438	0.8721	0.0159	0.8060	0.8402	0.0368	0.8613	0.8724
DAFNet	0.0125	0.9174	0.9191	0.0060	0.8922	0.9167	0.0325	0.8751	0.8841
GPNet	0.0112	0.9194	0.9255	0.0044	0.8885	0.9255	0.0321	0.8695	0.8845
GPNet-R	0.0092	0.9251	0.9284	0.0037	0.8934	0.9276	0.0334	0.8732	0.8879

As is shown, without any postprocessing, our method can highlight the salient object with better uniformity, and the interference of complicated backgrounds can be suppressed effectively.

2) Quantitative Comparisons: We conduct comparisons of the proposed method with other compared methods on three datasets. The comparison results are shown in Fig. 4 and Table I. Fig. 4 shows the PR curves, and Table I tabulates the results for the MAE, F-measure, and S-measure metrics. Fig. 4 shows that the proposed method achieves the best performance on the ORSSD dataset and competitive performance with DAFNet on the EORSSD and ORSI-4199 datasets. In terms of the MAE, the proposed method outperforms all the compared methods. The proposed method outperforms

TABLE II

F-MEASURE EVALUATION BASED ON THE SIZE ATTRIBUTE ON THE EORSSD DATASET

Method	£ 1%	1~10%	10~20%	20~30%	30~40%	³ 40%
DAFNet	0.8692	0.9233	0.9403	0.9428	0.9519	0.9633
Ours	0.9002	0.9352	0.9410	0.9418	0.9458	0.9551

the second-best method DAFNet by approximately 2% on the ORSSD dataset in terms of the F-measure. Nevertheless, it achieves competitive performance compared to DAFNet on the EORSSD and ORSI-4199 datasets. In terms of the S-measure, the proposed GPNet exceeds DAFNet by approximately 0.7% on the ORSSD dataset, approximately 1% on the EORSSD dataset, and approximately 0.001% on the ORSI-4199 dataset. In practice, the compared methods employ different backbones, such as VGG16 [57] and ResNeXt [58]. Therefore, we also provide the results of the proposed method under the superior ResNet backbone [59] (i.e., GPNet-R) for a relatively fair comparison in Table I. The proposed method using the ResNet backbone achieves competitive performance compared to DAFNet on the ORSI-4199 dataset and outperforms all the competitors on the ORSSD and EORSSD datasets, which demonstrates that the proposed method performs favorably compared to other compared methods.

The increase in salient objects and tiny objects usually makes detection challenging. Therefore, quantity- and size-based experiments are conducted to certify the advantages of the proposed method. Tables II and III show that our method achieves the best performance compared to the previous best methods in the case of multiple and tiny objects.

Moreover, to test our method's model size and speed, we compared it with the other five state-of-the-art methods. The results shown in Table IV validate that the proposed method outperforms other compared methods in model size

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TABLE III

F-MEASURE EVALUATION BASED ON THE QUANTITY ATTRIBUTE ON THE EORSSD DATASET

Method	0	1	2	3	4	5
DAFNet	0.9474	0.9170	0.8746	0.8886	0.9055	0.8848
Ours	0.9214	0.9257	0.8720	0.8915	0.9067	0.8972

TABLE IV

COMPARISON OF MODEL SIZE (MB) AND THE AVERAGE RUNNING TIME (SECONDS PER IMAGE) ON THE EORSSD DATASET

	LVNet	R3Net	R2Net	MINNet	DAFNet	Ours
Model size	207	148	172	184	112	76
Test time	0.74	0.48	0.59	0.62	0.36	0.29

TABLE V

EVALUATION COMPARISONS OF KEY COMPONENT ON THE EORSSD DATASET, I.E., AAM, FDS, AND GPM

Module	MAE ↓	F_b ↑	S_m ↑
Baseline	0.0235	0.7516	0.7912
Baseline + AAM	0.0164	0.8184	0.8645
Baseline + AAM + FDS	0.0086	0.8751	0.9178
Baseline + AAM + FDS + GPM	0.0044	0.8885	0.9255

and speed. Regarding the test time, our method only takes 0.29 s to generate a salient map, which is very efficient compared with the other five methods. All visual examples and quantitative metrics demonstrate the superiority and efficiency of the proposed method.

Moreover, the loss curves of the proposed GPNet over different datasets are shown in Fig. 5. It can be seen that all the loss curves decline rapidly in the first 500 epochs and level off after 1000 epochs. Thus, the training epoch is set to 2000 for better optimization.

E. Ablation Analysis

The proposed methods are mutually intensified to improve saliency object detection performance. In this section, we conduct ablation studies to analyze the effectiveness of the proposed methods (i.e., the AAM, FDS, and GPM)

1) Effectiveness of AAM: The AAM is designed to capture long-range dependencies and channel relationships for better feature representation. Table V shows that the baseline model with AAM achieves better performance than the baseline model. The performance is improved by 8.9% in terms of the *F*-measure, 13.1% in terms of the *S*-measure, and 30.2% in terms of the MAE. It is worth noting that the AAM does not need additional parameters, which indicates the effectiveness of the proposed AAM. In addition, we visualize the saliency map generated by the baseline model using the AAM. Fig. 6 clearly shows that the baseline module with the AAM generates better semantic similarity.

2) Effectiveness of FDS: To effectively encode the different levels of information, the FDS is constructed. Fig. 6 shows that the performance improvement of the baseline model with the FDS and AAM is obvious compared to the baseline with AAM, achieving 47.6%, 6.9%, and 6.2% improvements in terms of the MAE, *F*-measure, and *S*-measure, respectively. The same conclusion can be obtained from Fig. 6, i.e., the response of salient objects after installing the FDS is more accurate than that of the baseline with the AAM.

3) Effectiveness of GPM: The GPM is introduced to capture the relationships between all data samples. The results of the proposed model are reported in Table VI. The table shows that the proposed model achieves a performance improvement compared to the baseline with the AAM and FDS. The *F*-measure and *S*-measure increase by 1.5% and 0.8%, respectively. The visual effects of the GPM are shown in Fig. 6. The figure shows that the GPM helps effectively suppress the interference of backgrounds. Moreover, the comparison results of the feature masks are shown in Fig. 7. The figure shows

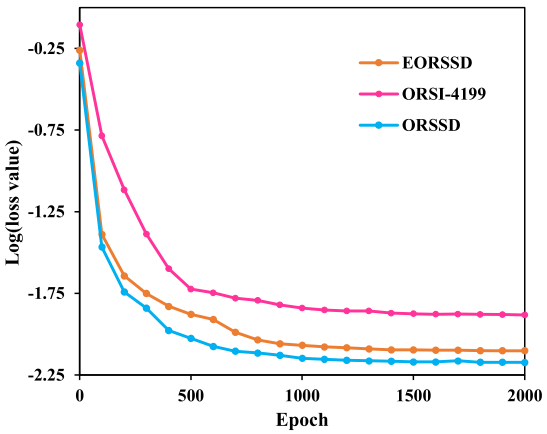


Fig. 5. Training loss of the proposed GPNet over different datasets.

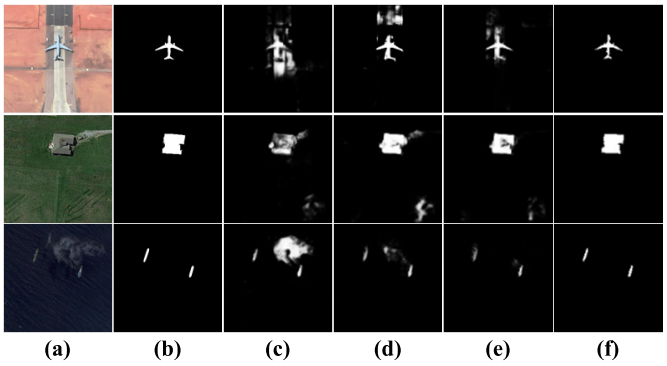


Fig. 6. Results of the comparisons showing the benefits of the proposed methods. (a) Images. (b) Ground truth. (c) Results of baseline. (d) Results of baseline with AAM. (e) Results of baseline with AAM and FDS. (f) Results of PGNet.

TABLE VI

F-MEASURE AND COMPUTATIONAL COMPLEXITY (GFLOPS) COMPARISONS WITH DIFFERENT ATTENTION MECHANISMS, I.E., SQUEEZE-AND-EXCITATION (SE) ATTENTION, DUAL ATTENTION, AND CBAM

Method	ORSSD	EORSSD	ORSI-4199	Computational complexity
Ours	0.9194	0.8885	0.8695	9.45
SE	0.9075	0.8757	0.8572	9.26
Dual attention	0.9185	0.8879	0.8686	9.61
CBAM	0.9104	0.8762	0.8594	9.26

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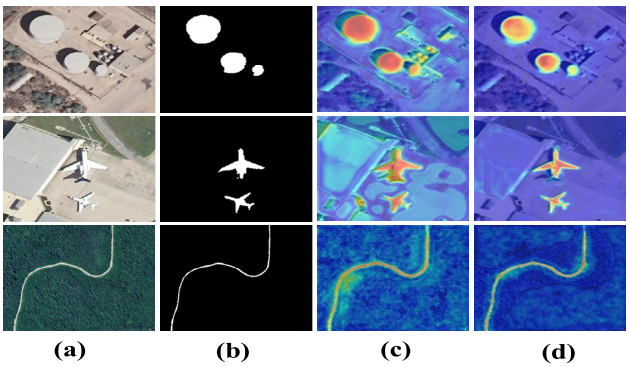


Fig. 7. Comparisons of the ablation study. (a) Images. (b) Ground truth. (c) Feature maps generated by the baseline. (d) Feature maps generated by PGNet.

that our proposed model effectively alleviates the influence of complex backgrounds and focuses on salient objects, which further demonstrates the effectiveness of the proposed GPM.

Moreover, comparative experiments of different attention mechanisms consisting of the squeeze-and-excitation (SE) attention [45], dual attention [46], and CBAM [48] are also conducted to further demonstrate the effectiveness of the proposed attention mechanisms. The comparative results of different attention mechanisms are reported in Table VI. It can be found that our proposed method achieves the best performance on all three datasets. Compared to the dual attention mechanisms, our proposed method achieves better performance and lower computational complexity. The reason is that the proposed GPM can capture the relationships between the entire dataset, and the AAB calculates the spatial- and channel-wise dependencies along the horizontal coordinate and the vertical coordinate, hence achieving performance improvement and lower computational complexity. Compared to the SE attention and CBAM, our proposed method achieves relatively considerable performance improvement and competitive performance of computation complexity. It demonstrates the effectiveness of the proposed attention mechanisms.

F. Deep Insight Into the Proposed GPNet

For clear description, the input and output forms of the proposed GPNet are reported in Table VII, and the feature visualizations of the proposed model are shown in Fig. 8. As shown, the feature representations in the encoder stage become increasingly more abstract due to the downsampling. Furthermore, the salient object gradually becomes explicit in the decoder stage.

Moreover, the GPM is proposed to capture the relationships between the entire datasets. The AAB is used to capture spatial- and channel-wise dependencies in one sample. Then, the global and the local attention information are mutually fused to enhance the network mode. The FDS is proposed to integrate information of different levels and filter the redundant information in different levels. In order to demonstrate the advantages and importance of the proposed methods, the feature maps generated after the proposed module are visualized in Fig. 9. The figure shows that the feature map generated after the GPM has more detailed features compared to Fig. 9(b), which demonstrates the effectiveness of the GPM. The feature map generated after the AAM [Fig. 9(e)] and FDS [Fig. 9(f)] contain better structural integrity of the detected object.

TABLE VII
INPUT AND OUTPUT FORMS OF THE PROPOSED MODEL

Layer		Input Size	Output Size
C1		(3, 128, 128)	(64, 128, 128)
GRM		(64, 128, 128)	(64, 128, 128)
C2		(64, 128, 128)	(128, 64, 64)
AAB1		(128, 64, 64)	(128, 64, 64)
	FDM1	(64, 128, 128), (128, 64, 64)	(128, 64, 64)
FSM1		(128, 64, 64), (128, 64, 64)	(128, 64, 64)
C3		(128, 64, 64)	(256, 32, 32)
AAB2		(256, 32, 32)	(256, 32, 32)
	FDM2	(128, 64, 64), (256, 32, 32)	(256, 32, 32)
FSM2		(256, 32, 32), (256, 32, 32)	(256, 32, 32)
C4		(256, 32, 32)	(512, 16, 16)
AAB3		(512, 16, 16)	(512, 16, 16)
	FDM3	(256, 32, 32), (512, 16, 16)	(512, 16, 16)
FSM3		(512, 16, 16), (512, 16, 16)	(512, 16, 16)
C5		(512, 16, 16)	(512, 8, 8)
AAB4		(512, 8, 8)	(512, 8, 8)
	FDM4	(512, 16, 16),(512, 8, 8)	(512, 8, 8)
FSM4		(512, 8, 8), (512, 8, 8)	(512, 8, 8)
D4		(512, 8, 8), (512, 16, 16)	(256,16,16)
D3		(256,16,16), (256, 32, 32)	(128, 32, 32)
D2		(128, 32, 32), (128,64,64)	(64,64,64)
D1		(64,64,64), (64,128,128)	(1, 128, 128)

TABLE VIII
SENSITIVITY ANALYSIS OF THE GPNET OVER THE LEARNING RATE, BATCH SIZE, AND INITIAL WEIGHT VALUES

	Parameter	ORSSD	EORSSD	ORSI-4199
Learning Rate	0.0002	0.9184	0.8867	0.8691
	0.00015	0.9188	0.8879	0.8695
	0.0001	0.9191	0.8885	0.8675
	0.00005	0.9162	0.8858	0.8604
Batch Size	16	0.9186	0.8881	0.8706
	8	0.9194	0.8885	0.8695
	4	0.9164	0.8876	0.8659
Initial Weight Value	Xaiver	0.9187	0.8891	0.8667
	Kaiming	0.9191	0.8885	0.8695

G. Sensitivity Analysis of Parameters

To demonstrate the robustness of the proposed GPNet, the experiments with different parameters are conducted. The experiments of different parameters consisting of the learning rate, batch size, and initial weight values are performed on three datasets. The learning rate is initialized to 1×10^{-4}, 1.5×10^{-4}, and 5×10^{-5}. Due to the limitation of GPU memory, the batch size is set to 16, 8, and 4. To analyze the proposed GPNet's sensitivity of initial weight value, methods consisting of Xaiver and Kaiming are selected. For a fair comparison, all other parameters remain unchanged. As reported in Table VIII, the experimental results show that the parameters, including learning rate, batch size, and initial weight value, do not significantly affect the network performance, which

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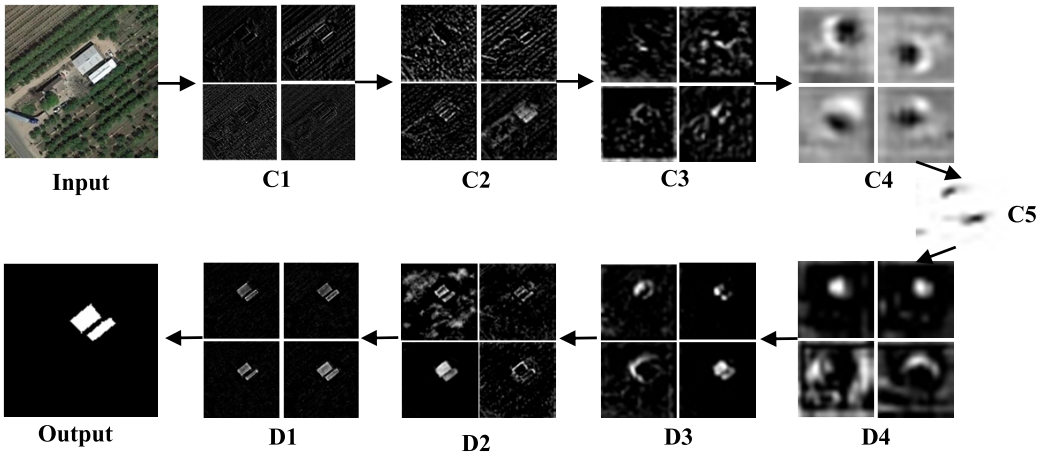


Fig. 8. Feature visualization of the proposed GPNet.

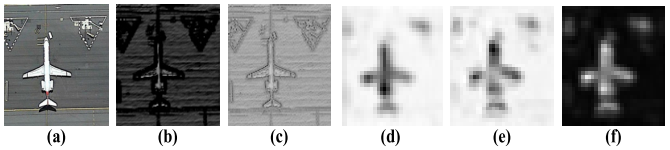


Fig. 9. Feature comparisons of different modules. (a) Images. (b) Results of first backbone block (C1). (c) Result of first backbone block with global attention module. (d) Result of second backbone block (C2). (e) Result of second backbone block with AAM. (f) Result of second backbone block with FDS.

TABLE IX
F-MEASURE EVALUATION WITH THE TRAINING RATIOS OF 50% AND 80% ON THREE DATASETS

Dataset	Method	50% Training Ratio	80% Training Ratio
ORSSD	Ours	0.7483	0.8568
	DAFNet	0.7421	0.8497
EORSSD	Ours	0.7794	0.8471
	DAFNet	0.7752	0.8462
ORSI-4199	Ours	0.8046	0.8597
	DAFNet	0.8127	0.8572

demonstrates that the proposed GPNet is not sensitive to these parameters when set in a reasonable range.

H. Analysis of Different Training Ratios

To make a comprehensive evaluation, comparative experiments of different training ratios are conducted. Specifically, the training ratios for all three datasets are set to 50% and 80%. The comparative results of different training ratios are reposted in Table IX. As can be seen, our proposed model achieves the best results on all three datasets with the training ratio of 80%. Under the training ratio of 50%, our proposed GPNet achieves the best results on the ORSSD and EORSSD datasets and competitive performance compared to the previous best method (i.e., DAFNet) on the ORSD-4199 dataset. It demonstrates the effectiveness and robustness of the proposed GPNet.

V. CONCLUSION

In this study, a GPNet is constructed for salient object detection from optical remote sensing images. First, a GPM is designed to capture the global attention relationships among the entire dataset. Then, an AAB is proposed to obtain both spatial and channel dependencies. Moreover, an FDS is introduced to integrate the multilevel features sequentially flowing. Through this structure, the global attention relationships from the GPM are transformed to every level to guide feature generation. The interference of the complex backgrounds and scale variation can be effectively alleviated using the proposed methods. Comprehensive experiments demonstrate the effectiveness and robustness of the proposed GPNet. In future work, designing a more robust network by using much less data for training is a topic worthy of further investigation.

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